



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	Intelligent AI-Based Inventory Management with Predictive Stock Analysis	
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Abstract (batch no:-05)

Intelligent AI-Based Inventory Management with Predictive Stock Analysis is an adaptive software system designed to automate and optimize inventory operations using **Artificial Intelligence and Machine Learning** techniques. The system manages product stock, supplier information, sales records, and inventory levels while analyzing historical sales and inventory data to predict future demand. Based on these predictions, the system identifies potential **stock shortages and overstocking**, generates timely stock alerts, and recommends suitable reorder quantities. It provides real-time inventory monitoring and intelligent stock recommendations, helping businesses maintain optimal inventory levels and make better purchasing decisions. The system continuously learns from changing sales patterns and newly available data, allowing its predictions and recommendations to adapt to variations in customer demand. This adaptive approach reduces manual inventory monitoring, minimizes stock-related losses and unnecessary inventory costs, and improves overall operational efficiency. Thus, the proposed system provides a **data-driven, intelligent, adaptive, and cost-effective solution for modern inventory management**.

Signature of the Guide

HOD